

National University of Computer & Emerging Sciences
Karachi Campus



**BUS TICKET RESERVATION AND INFORMATION
MANAGEMENT SYSTEM**

Project Proposal
Programming Fundamentals
Section: C

Group Members:
26k-0608 Okasha Faisal
26k-0603 Mohammad Bilal Ahmed
26k-0588 Muhammad Aarij Khurram

Project Proposal

Bus Ticket Reservation and Information Management System

Introduction

The Bus Ticket Reservation and Information Management System is an application that will provide a simple way to manage bus, passenger, seat, and reservation information.

The system will allow users to view available buses, search for them according to their routes, check seat availability, reserve tickets, cancel reservations, and view reservation details. This project will demonstrate the practical application of pf concepts in C to solve a real-world information management problem.

Existing System

Bus ticket reservations are commonly handled through booking counters and online reservation systems. These systems allow passengers to select a route, bus, departure time, and seat before making a reservation.

Commercial bus reservation systems generally contain many advanced features and complex interfaces. For Programming Fundamentals project, a simpler information management system is more appropriate. A basic reservation process still requires proper management of bus information, passenger details, seat availability, and reservation records.

[An existing system for reference](#)(Faisal Movers)

Problem Statement

In a manual or basic bus reservation process, managing bus schedules, passenger information, and seat availability can become difficult. Manual handling of reservations may also result in problems such as difficulty finding suitable buses, lack of immediate information about available seats, assigning an already reserved seat, and difficulty retrieving or cancelling an existing reservation.

Our system aims to address these problems by providing a console-based system for managing bus and reservation information. The system will introduce organized bus searching, seat availability checking, reservation management, and reservation cancellation.

Proposed Solution

We propose to develop a console-based Bus Ticket Reservation and Information Management System using the C programming language. The system will maintain information about buses, routes, seats, passengers, and reservations.

Users will be able to view and search available buses, check the availability of seats, reserve a selected seat, cancel a reservation, and view reservation details. The system will also include input validation to reduce invalid operations and prevent users from reserving an already occupied seat.

The application will use appropriate C programming concepts such as variables, conditional statements, loops, arrays, functions, strings, structures, and file handling.

Salient Features

- View available buses and their information
- Search buses according to departure and destination
- Display seat availability
- Reserve a bus ticket
- Enter and manage passenger information
- Generate a reservation ID
- View reservation details
- Cancel an existing reservation
- Prevent reservation of already occupied seats
- Validate user input
- Store and retrieve reservation information
- Menu-driven console interface

Tools & Technologies

Programming Language: C

Compiler: GCC

Framework: Not Applicable

Version Control: Git and GitHub

Application Type: Console-Based Application

Operating System: Windows

IDE: Visual Studio Code